Simple Pendulum Motion

Introductory physics · ~2 pages · for XR EduAgent study

Learning goals

After this short reading you should be able to: (1) state what a simple pendulum is; (2) recall the small-angle period formula; (3) explain how length and gravity affect period; (4) distinguish amplitude from period for small angles.

What is a simple pendulum?

A simple pendulum is a point mass (bob) attached to a light inextensible string of length L, swinging under gravity about a fixed pivot. Real classroom pendulums approximate this ideal.

Forces and restoring torque

When displaced by angle θ from vertical, the component of weight tangential to the arc is $-mg \sin\theta$. For small angles (θ in radians, $|\theta|$ ■ 10–15°), $\sin\theta \approx \theta$, so the motion is approximately simple harmonic with angular frequency $\omega = \sqrt{(g/L)}$.

Period formula (small angle)

The period T (time for one full swing) is $T = 2\pi \sqrt{(L/g)}$. Key consequences: longer L $\rightarrow$ larger T; stronger g $\rightarrow$ smaller T; for small angles, T is approximately independent of amplitude.

Common misconceptions

Students often think heavier bobs swing faster. In the simple-pendulum model, mass cancels and does not appear in T. Large amplitudes break the small-angle approximation and lengthen the real period slightly.

Why 3D / interactive learning helps

Seeing length change while a live period readout updates, and comparing two pendulums side by side, makes the $\sqrt{L}$ dependence and the mass-independence claim concrete.